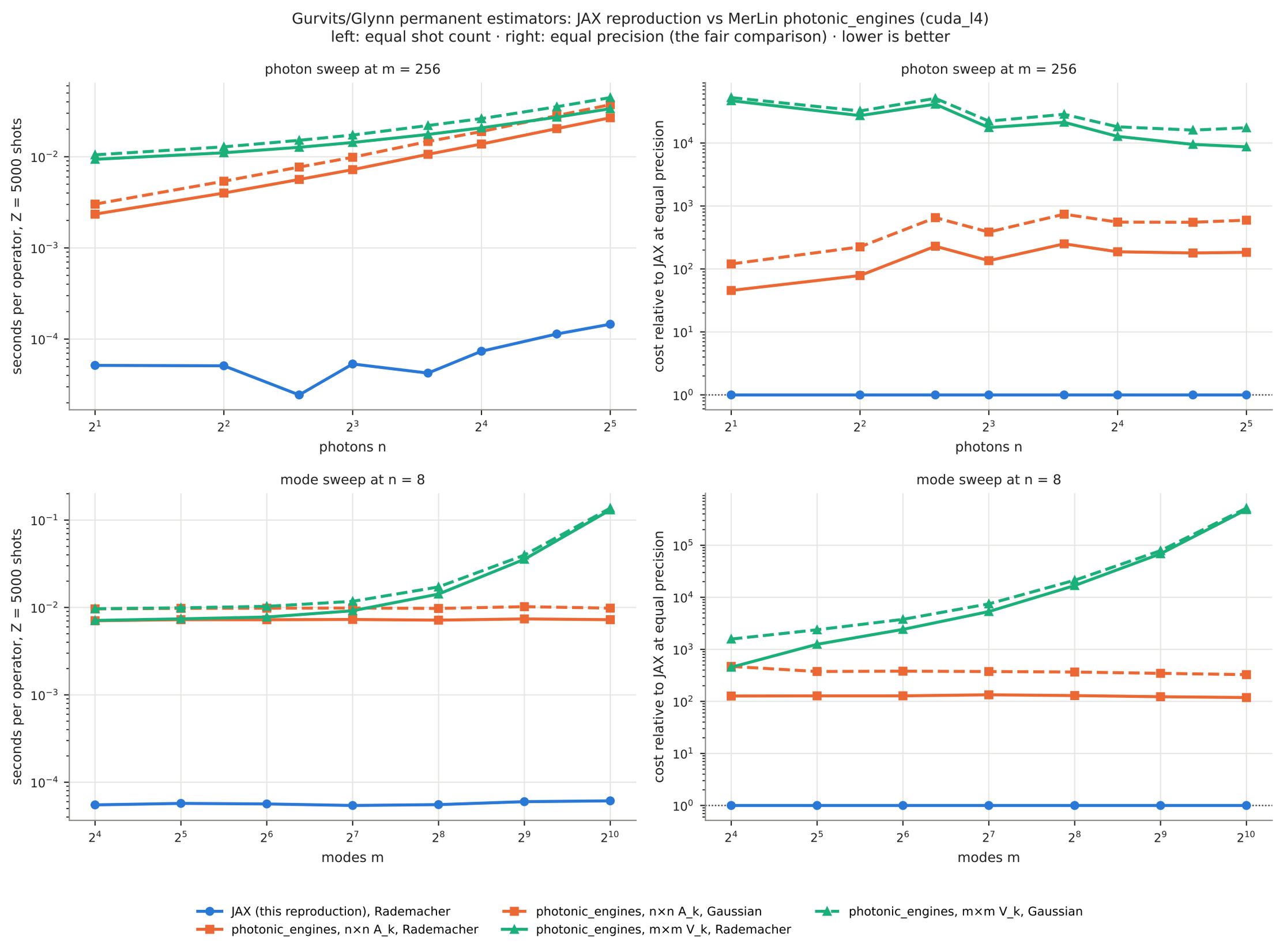
Gurvits/Glynn permanent estimators: JAX reproduction vs MerLin photonic_engines (cuda_l4)
left: equal shot count · right: equal precision (the fair comparison) · lower is better
photon sweep at m = 256
seconds per operator, Z = 5000 shots
photons n
photon sweep at m = 256
cost relative to JAX at equal precision
photons n
mode sweep at n = 8
seconds per operator, Z = 5000 shots
modes m
mode sweep at n = 8
cost relative to JAX at equal precision
modes m
JAX (this reproduction), Rademacher
photonic_engines, n×n A_k, Rademacher
photonic_engines, n×n A_k, Gaussian
photonic_engines, m×m V_k, Rademacher
photonic_engines, m×m V_k, Gaussian